

NATALIA KONG YI

Product Designer · Systems Thinker · Design Researcher

Translating complexity into meaningful digital experiences

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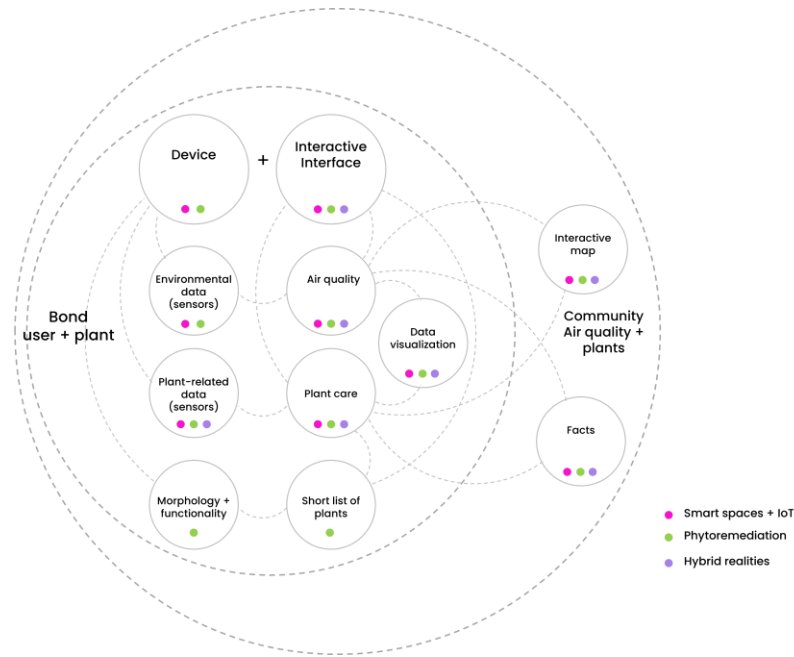
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I design experiences that help people understand, navigate, and shape complex systems.

My practice combines design research, systems thinking, and experience design to explore how technology, services, and environments influence the way people interact and make decisions.

The projects in this portfolio span healthcare, logistics, architecture, creative technology, and speculative design. Across these contexts, I approach products and experiences as interconnected ecosystems — designing solutions that connect people, information, and technology through research, prototyping, and collaboration across disciplines.

PRACTICE



Conceptual system map. *Relationships between environmental sensing, digital representation, and user interaction, used to align the platform's architectural logic with the intended user experience.*

HYPERPLANT

Designing hybrid experiences that cultivate long-term relationships between people and living systems

Role: Interaction & Systems Designer · Creative Technologist

Type: Master's Thesis · Research through Design

Year: 2024

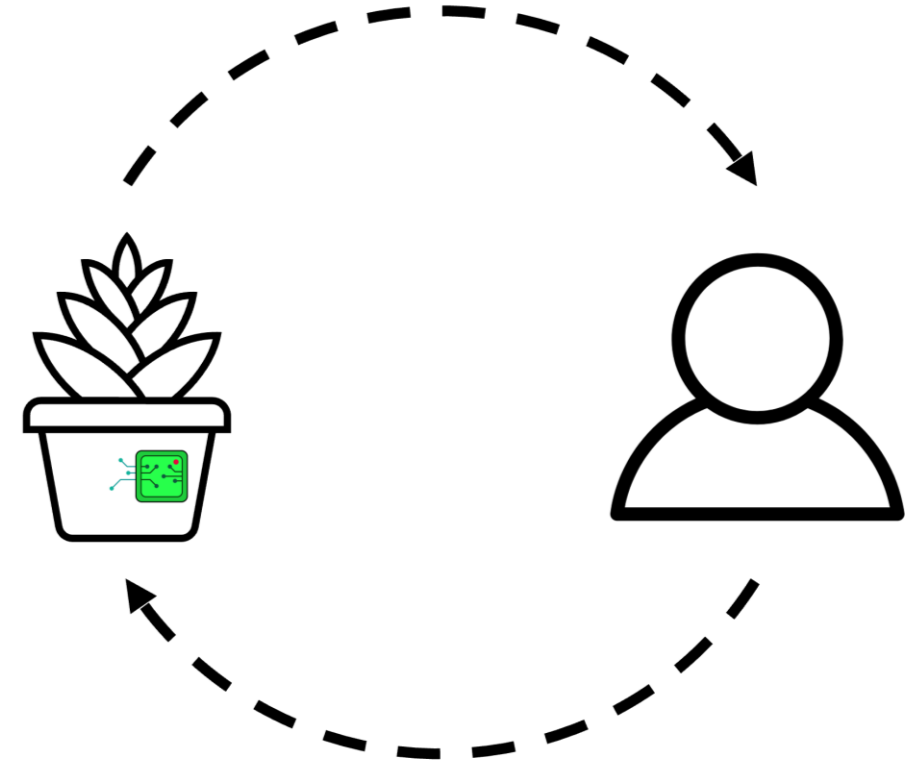
Team: Individual Project

CHALLENGE

Current models of production and consumption often separate people from the ecological systems that sustain them. Although environmental data can make these processes visible, it rarely fosters lasting awareness or behavioral change.

How might hybrid interactive systems cultivate care and responsibility toward living systems?

Research-through-design project that aims to reframe human-nature relationship through interaction and data. HyperPlant integrates living digital twins, environmental sensing, and interactive feedback into a hybrid ecosystem that extends human perception beyond observation toward ongoing engagement.



Conceptual Framework. Defining the platform architecture and the relationship between sensing and user interaction.

OUTCOMES

Developed and evaluated a hybrid system combining IoT, interaction, and behavioral design to explore long-term engagement with living systems.

Hybrid Interaction Ecosystem

Integrated environmental sensing, physical computing, and a web platform into a connected physical–digital experience.

Living Digital Twins

Translated environmental conditions into expressive digital representations that made ecological change perceptible through everyday interaction.

Smart Plant Wearable

Designed and prototyped a 3D-printed wearable that monitors environmental conditions while supporting air phytoremediation.

Community Platform

Proposed a digital platform for monitoring, storytelling, and future community participation.

Behavioral Evaluation

Conducted a one-week user evaluation using participant diaries and the Octalysis framework to examine engagement and motivational patterns.



System Components. The underlying components required to capture, process, and transmit environmental data.

TAKEAWAYS

- Invisible systems become meaningful through interaction. Design can translate imperceptible ecological processes into everyday experiences.
- Reciprocity strengthens care. Making mutual influence visible fosters more meaningful relationships with living systems.
- Hybrid experiences connect perception and action. Physical and digital interactions reinforce one another when grounded in the same reality.
- Narrative sustains engagement. Stories and feedback turn awareness into sustained action.
- Embodied technologies expand perception. Sensors, computation, and biology create new ways of perceiving and interacting with the living world and environment.

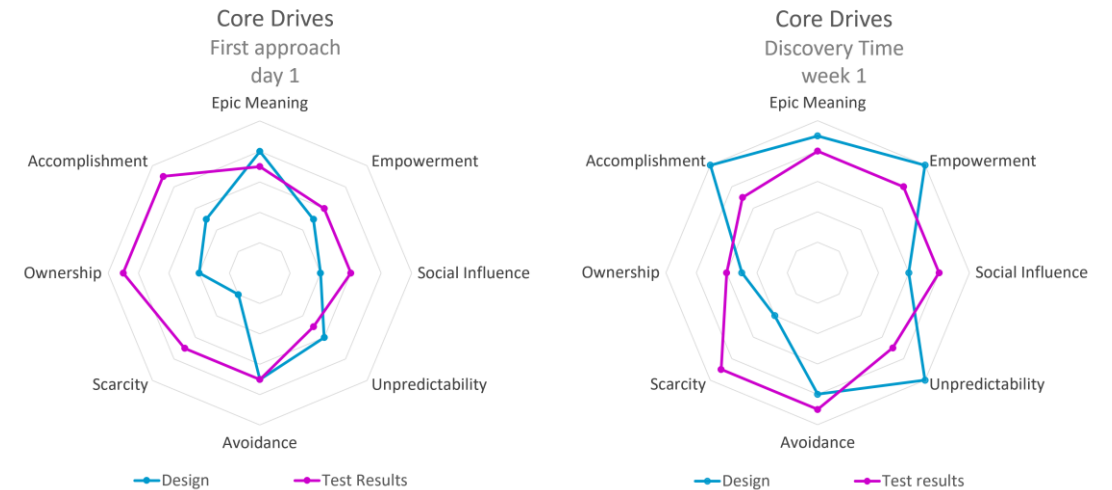
Methods: Research-through-design · Comparative research · Physical prototyping · AI-assisted coding · one-week user evaluation · Octalysis analysis

Tools: Arduino · Sensors · Physical computing · 3D printing · Interactive prototyping

Presented at: Chinese Conference on Bio-Inspired Design and Technology (2020)

· International Image Festival (Colombia, 2021)

More: [Full case study](#) →



Behavioral Analysis. Using the Octalysis framework to measure the alignment between designed motivation models and real user engagement.



Beyond the Clinic. *Conceptual framework used to understand the relationships between patients, knowledge, community, and care services before defining individual interventions.*

DIABETES INTEGRAL HEALTHCARE CENTER

Designing connected care experiences across fragmented healthcare services

Role: Impact Designer · Service Design Team

Year: 2022

Type: Service Design · Healthcare Innovation · Participatory Design

Team: Viernes Peru · Multidisciplinary design team

CHALLENGE

Diabetes management extends beyond clinical care, reshaping everyday routines, relationships, and long-term lifestyle change. Yet most healthcare services remain focused on medical treatment rather than the broader support systems that sustain it.

How can healthcare services support the everyday relationships and practices that influence long-term treatment engagement?

Through participatory research, the team explored how care could extend beyond clinical interactions by designing services, tools, and experiences that foster empathy, dialogue, and shared responsibility among patients and their support networks.



Ease
Burden



Share
Experiences



Encourage
Reflection



Actionable
Knowledge



Spark
Curiosity

Learning principles. *Design principles used to guide every learning artifact and experience across the project.*

OUTCOMES

Developed a service ecosystem integrating research insights, interaction tools, and strategic frameworks to support long-term diabetes care.

Care Ecosystem

Reframed diabetes management as an ecosystem shaped by relationships, routines, and everyday contexts rather than clinical interactions alone.

Shared Practices

Designed and prototyped interaction tools—including conversation cards and an interactive installation—to encourage empathy and collaborative care. Experience prototype evaluated with 10 participants.

Service Strategy

Created a design playbook translating research insights into reusable principles for future healthcare services.

Learning Artifact Design Canvas

Subject  1 What topic are we exploring? What core idea do we want to develop? How does this idea contribute to the learning and community ecosystem?	Objectives:  2 How does this topic support the strategic goals of the healthcare center? What outcomes are we aiming to achieve? Who is this experience designed for? What insights or outputs do we hope to generate?	Design & Fabrication:  7 How will this artifact be designed? What are its key components? What format will it take? Does it require any supporting elements for implementation (furniture, lighting or acoustic systems, audiovisual equipment, or other resources)?
Relational system:  4 How does this artifact connect with the broader learning ecosystem? What other learning artifacts or community activities does it relate to? What upcoming community events should it support or build upon?	Approach:  3 How should this topic be approached? In what context will this artifact be experienced (public, semi-private, or private)? What level of participation or visibility should it encourage?	
Principles:  5 How should the learning principles be reflected in this artifact? Rate the relevance of each principle and indicate its importance using the corresponding scale.	Users:  6 Who is this artifact designed for? What experience, behavior, or outcome do we want to encourage for each audience?	

Learning Artifact Design Canvas. *Framework for translating strategic objectives into cohesive learning experiences.*

TAKEAWAYS

- Care extends beyond the clinic. Health outcomes are shaped by relationships, routines, and everyday environments as much as by clinical interventions.
- Relationships are part of the care system. Designing opportunities for dialogue and shared reflection can strengthen long-term treatment engagement.
- Design frameworks enable continuity beyond individual interventions. Translating research into reusable principles enables teams to design, evaluate, and evolve services beyond a single intervention.

Research & Design: 6 stakeholder interviews · Service ecosystem mapping · Co-design workshops · MVP validated with 10 patients

Outputs: Ecosystem map · Learning Playbook · Family Playbook · Design canvases



Water Flow Series. *Exploring rivers as living data streams, where environmental conditions become visual narratives.*

HYBRID NATURES

Designing participatory infrastructures for engaging with living systems through hybrid experiences.

Role: Strategy & Systems Designer · Design Research · Creative Technologist

Year: 2025

Type: Master's Thesis (Creative Computing, ELISAVA / UVic-UCC)

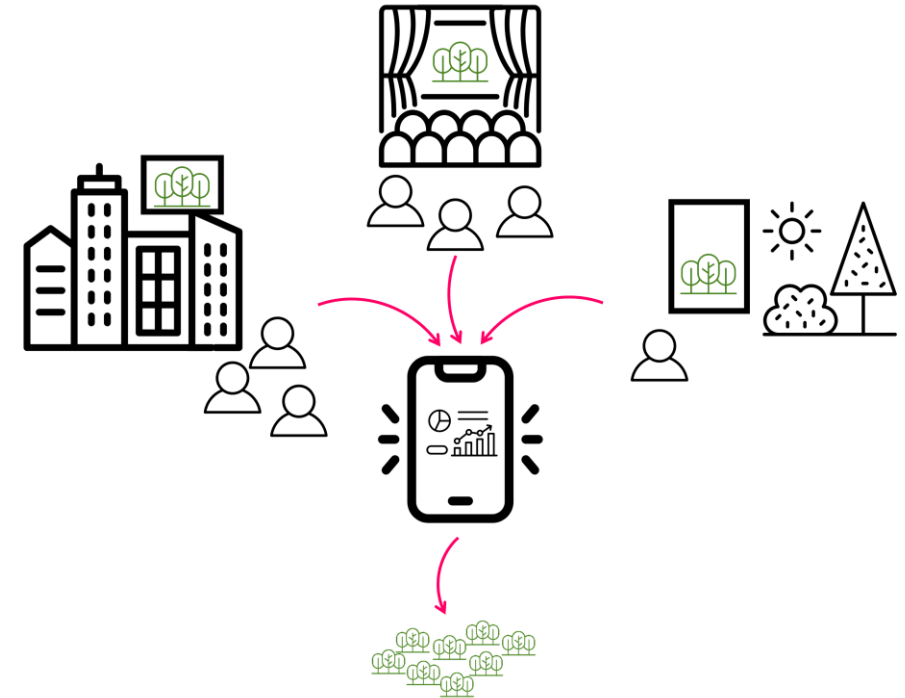
Team: Independent project

CHALLENGE

Hybrid Natures is a research-through-design project exploring how hybrid physical–digital systems can help people understand and engage with complex socio-environmental systems. Using the Amazon Rainforest as a case study, it investigates how data, narrative, and interaction can make distant ecosystems perceptible through participatory experiences.

How can design and technology make distant human–nature relationships legible and actionable at a human scale?

Rather than designing a standalone product, I approached the challenge as a systems problem, integrating interaction, governance, and participation into a single experience.



Hybrid Natures Concept. Conceptual framework connecting environmental data, digital ownership, and collective stewardship through a decentralized platform.

OUTCOMES

Developed a proof of concept integrating interaction design, systems thinking, creative technology, and incentive models to explore new forms of collective environmental stewardship.

Narrative

Designed a data-driven experience that guides participation from awareness to contribution.

Participation

Integrated governance, digital ownership, and collective decision-making into a participatory system concept.

Value Model

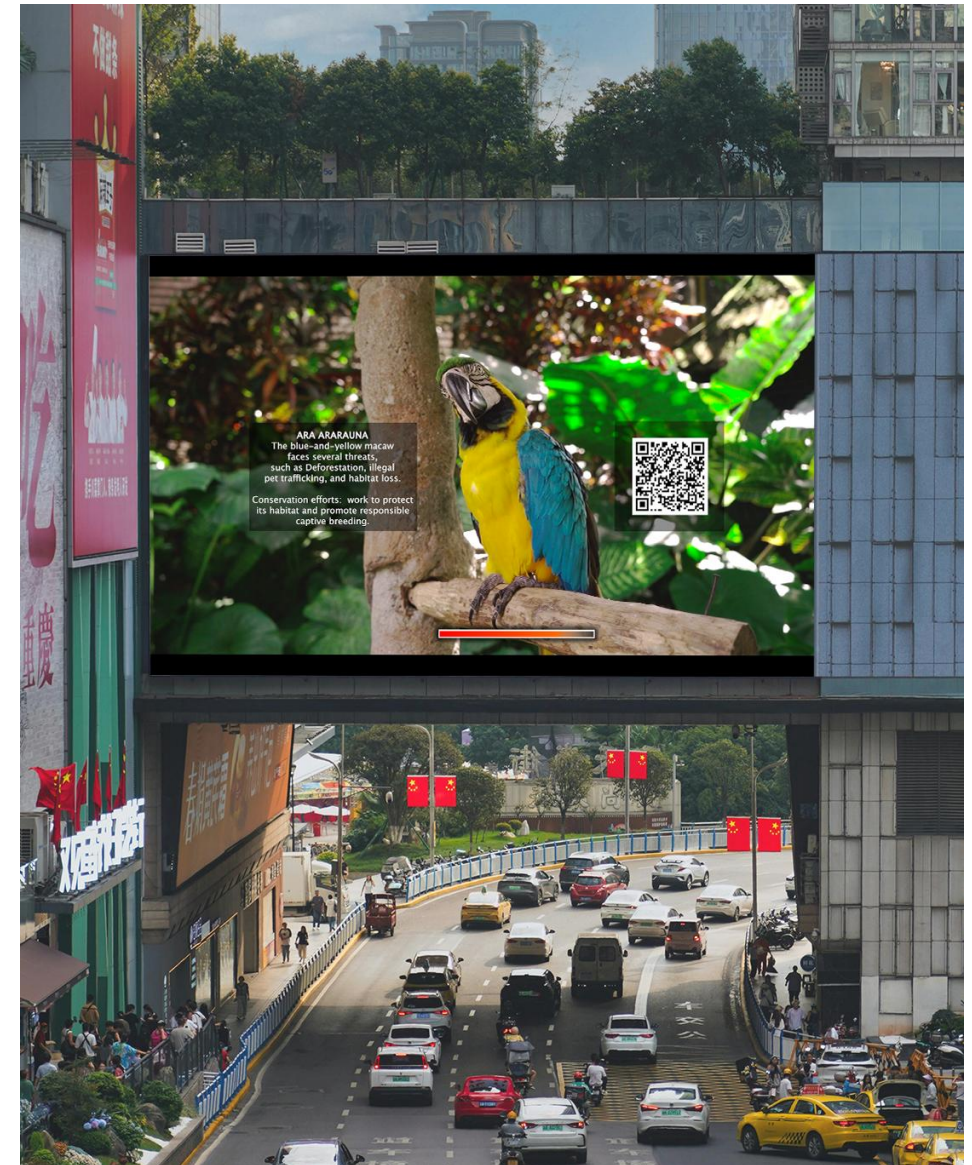
Proposed a shared-value model allocating revenue between artists (33%) and conservation NGOs (67%) to align economic and environmental incentives.

Hybrid Infrastructure

Defined a platform architecture connecting digital participation with distributed physical touchpoints.

Creative Technology

Built a functional prototype using AI-assisted development, generative visualization, and web technologies.



Decentralized screens (speculative). *Interface concepts for discovering environmental narratives and supporting conservation through digital ownership.*

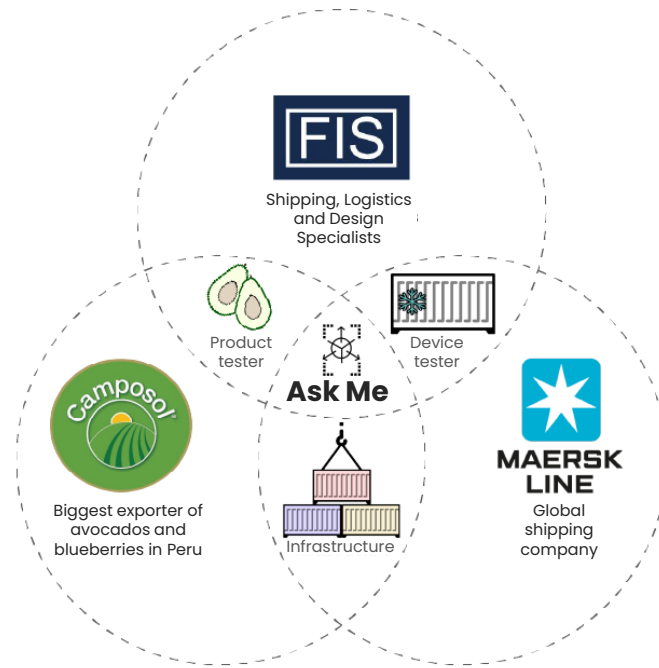
TAKEAWAYS

- Design participation intentionally. Long-term engagement emerges when ownership, contribution, and progression become part of the experience.
- Governance shapes the product experience. Decision-making and transparency influence how people engage—not only how systems operate.
- Narrative makes complexity navigable. Structuring information across multiple scales helps people build mental models of interconnected systems.
- Hybrid systems extend engagement. Physical and digital touchpoints reinforce one another, creating richer and more persistent participation.
- Speculation is a design method. Designing future scenarios makes assumptions tangible before implementation.

Tools: p5.js · HTML/CSS · Flourish · AI-assisted development

Presented at: International Colloquium Extractivismos (Argentina, 2025)

More: [Full case study](#) →



Collaboration Agreement. *Articulation between different stakeholders (producers and logistics) for innovation.*

ASK ME

Making invisible conditions visible in global cold-chain logistics

Role: Design Director · Systems Designer & Strategist

Type: Innovation Concept · IoT Exploration · Systems Design

Year: 2021

Team: FIS Team · Research Collaboration with Camposol, Maersk & PUCP

CHALLENGE

Cold-chain failures cause significant food loss before products reach consumers. While large exporters rely on monitoring systems and operational expertise, smaller producers often lack visibility into how perishable goods evolve during transportation.

How might logistics systems perceive and respond to the changing conditions of living cargo?

Building on previous research into living systems and IoT, ASK ME explored how sensing technologies could translate biological and environmental conditions into actionable information through an adaptive cargo monitoring concept.

OUTCOMES

Explored how IoT, systems thinking, and interaction design could make the condition of living cargo visible through an adaptive monitoring concept.

Adaptive Monitoring Concept

Proposed an IoT-based monitoring concept for revealing the evolving condition of perishable goods throughout transportation.

Cargo Data Representation

Conceptualized a data model translating biological and environmental signals into actionable logistics insights.

Industry Collaboration

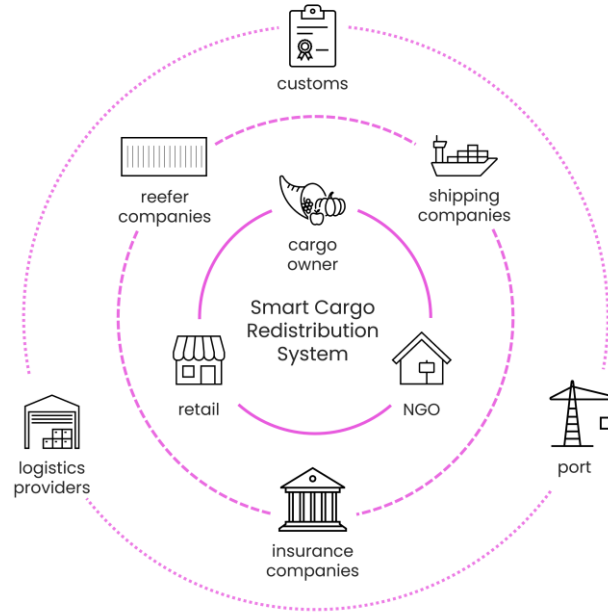
Established a research collaboration with Camposol, Maersk, and PUCP to explore future development and testing. Implementation was postponed due to the COVID-19 pandemic.

TAKEAWAYS

- Make invisible systems perceptible. Translating hidden processes into meaningful information supports better decisions.
- Design relationships, not isolated products. Technology creates value by connecting actors, data, and decisions.
- Explore before scaling. Early concepts align stakeholders and reveal opportunities before implementation.

Methods: Research-through-design · Systems thinking · IoT exploration · Interaction design

Scope: Cold-Chain Logistics · Food Systems · Adaptive Technologies



Smart Cargo Redistribution System. *Mapping the multi-lateral coordination between maritime IoT data, commercial decision-makers, and humanitarian end-nodes.*

SMART CARGO

Designing resilience into global maritime logistics

Role: Systems Designer & Strategist

Type: Independent Systems Strategy

Year: 2026

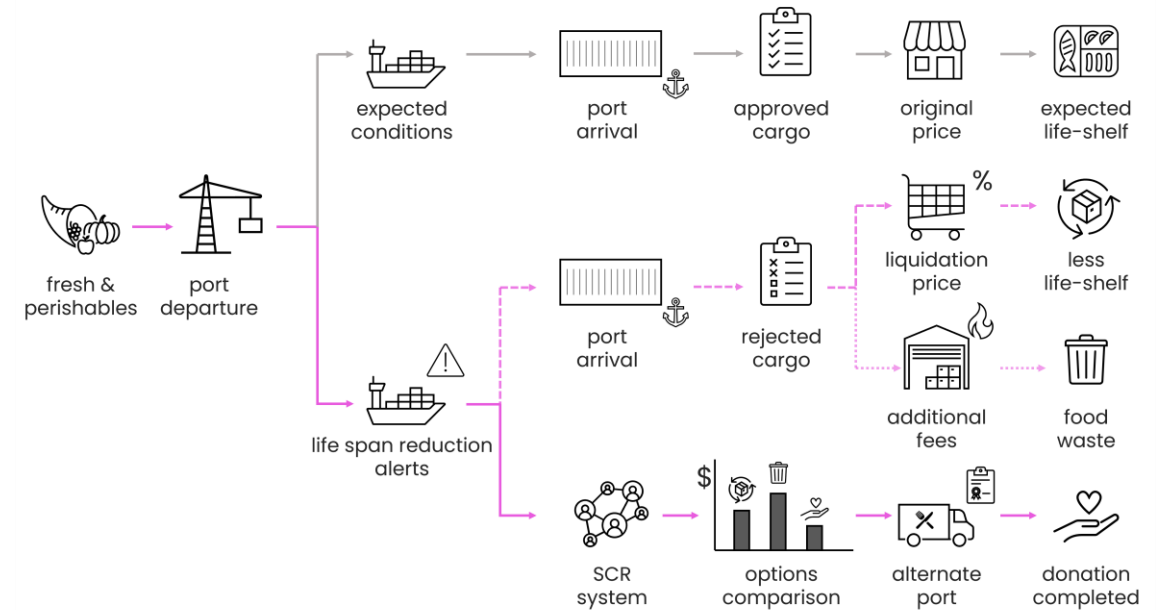
Team: Self-initiated

CHALLENGE

Cold-chain logistics depends on producers, exporters, carriers, and distributors making time-sensitive decisions. While large exporters often rely on monitoring systems and operational expertise, smaller producers have limited visibility into cargo conditions during transportation.

How might we transform logistical failures into humanitarian opportunities by making cargo rescue a more valuable alternative than disposal?

Instead of replacing existing logistics infrastructure, Smart Cargo explores a decision-support layer that uses existing IoT data to evaluate redistribution pathways before food integrity is compromised. The concept builds on previous logistics research and examines how interoperability can enable more resilient food systems.



Redistribution Logic Journey. *Comparative flow of the intervention journeys and outcomes.*

OUTCOMES

Since this is a strategic concept, I'd avoid language that implies software already exists.

Rescue Decision Framework

Proposed a strategic framework for evaluating cargo recovery against operational, economic, and humanitarian outcomes.

Redistribution Logic

Defined a coordination model connecting logistics providers, ports, NGOs, and food banks through alternative redistribution pathways.

Alternative Scenarios

Conceptualized a decision-support model comparing rescue strategies based on timing, cost, and humanitarian impact.

Interoperability Strategy

Proposed a software layer designed to integrate with existing maritime systems rather than replace them.

Circular Value Model

Explored how aligned business, environmental, and humanitarian incentives could make cargo rescue a viable alternative to disposal.

TAKEAWAYS

- Design shape behavior. Sustainable systems succeed when the preferred outcome is also the most valuable.
- Innovation can emerge through coordination. Connecting existing infrastructure often creates more value than replacing it..
- Decision support creates resilience. Strategic frameworks help transform operational failures into new opportunities..
- Interoperability enables ecosystems. Coordinating fragmented actors creates greater impact than optimizing isolated services.
- Shared value strengthens resilience. Aligned incentives help environmental, social, and business goals reinforce one another.

Methods: Systems Thinking · Service Design · Strategic Foresight

Scope: Humanitarian Logistics · Circular Economy · Decision Support Systems

More: [Full case study](#) →

Education

2024-2025 | Master in Creative Computing. ELISAVA & University of Vic – Central University of Catalonia.

2018-2019 | M. Des in Interactive Design. University of Buenos Aires. (Thesis defended in 2024)

2010 | B. Arch. Pontifical Catholic University of Peru

Research + Exploration Areas

Hybrid Experiences · Systems · Creative Computing · Participatory Models · Generative Design

Publications, Conferences and Lectures

2025 | International Colloquium ExtrActivismos, National University of San Martin. Project presentation (Hybrid Natures).

2022 | Design, Art and Science chair, University of Bogota Jorge Tadeo Lozano. Guest lecture & project presentation (HyperPlant).

2021 | EXPERIENCE DESIGN Korea & Latin America Research Exchange. Human and Design Press. Seoul (KR). Paper co-author (published).

2021 | International Image Festival (CO). Project presentation (HyperPlant) 2020 | Chinese Conference on Bio-inspired Design and Technology (CN). Project presentation (HyperPlant)

Selected Projects

2026 | Kuya Silver Mining Facilities — Architectural Consultant

2026 | Smart Cargo — Strategy Designer

2025 | Seniors Club — Strategy Designer

2025 | Zapallal Mall — Architectural Consultant

2024 | Talara Mall — Architectural Consultant

2024 | HorrorLand — Experience Designer (Immersive Systems)

2023 | Circurela — Designer (Pop-up & Circular Systems)

2022 | Diabetes Integrated Healthcare Center — Experience Designer

2022 | La Molina Mall — Architectural Consultant

2021 | AskMe — Design Director

2019 | S'Dely Corporate Building & Showroom — Principal Architect

2018 | MegaPlaza Resettlement — Architectural Consultant

2017 | Osinergmin — Design Standards Handbooks (National Scale)

2016 | MegaPlaza — Design Standards Handbooks (National Scale)

2015 | VeggieLab — Principal Architect

2014 | Onduline — Marketing & Technical Department Manager

2010 | Participatory Urban Design Workshop — Urban Consultant

EXPERIENCE